CASE STUDY 2 – PROGRESSIVE ELECTRICITY BILLING

1. Problem Statement

Calculate household electricity bills using consumption slabs, fixed charges, surcharges, and rebates; compare monthly bills and identify unusual increases.

2. Objectives

- Calculate energy charges using progressive slabs.
- Add fixed charges and a surcharge when applicable.
- Apply a rebate for low consumption.
- Compare monthly bills and flag an unusual increase.

3. Assumptions

- First 100 units cost ₹2.00/unit; next 100 cost ₹3.00/unit; units above 200 cost ₹5.00/unit.
- Fixed charge is ₹50. A 5% surcharge applies when energy charges exceed ₹500.
- A 5% rebate on energy charges applies when consumption is at most 100 units.
- An increase greater than 30% from the previous month is unusual.

4. Algorithm

- Store monthly unit readings in a list.
- Calculate slab-wise energy charges for each month.
- Apply rebate or surcharge, then add fixed charge.
- Compare each bill with the previous month's bill.
- Display each bill and flag increases above 30%.

5. Pseudocode

START

Store monthly consumption.

FOR each month: calculate slab charges.

Apply rebate if units <= 100; otherwise apply surcharge if energy charge > 500.

Add fixed charge and store total bill.

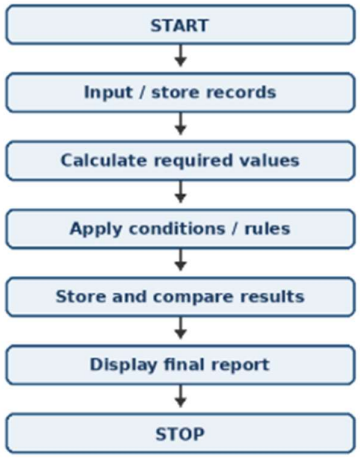
Compare current bill with previous bill and calculate percentage increase.

Display bills and unusual increases.

STOP

6. Flowchart

Process flow (high-level):



7. Python Source Code

```
monthly_units = [80, 120, 190, 280]
fixed_charge = 50
bills = []
for month in range(len(monthly_units)):
    units = monthly_units[month]
    if units <= 100:
        energy_charge = units * 2
    elif units <= 200:
        energy_charge = 100 * 2 + (units - 100) * 3
    else:
        energy_charge = 100 * 2 + 100 * 3 + (units - 200) * 5
```

```
        adjustment = 0
        if units <= 100:
            adjustment = -energy_charge * 0.05
        elif energy_charge > 500:
            adjustment = energy_charge * 0.05
        total_bill = energy_charge + adjustment + fixed_charge
        bills.append(total_bill)
        print(
            "Month", month + 1,
            "| Units:", units,
            "| Energy:", round(energy_charge, 2),
            "| Adjustment:", round(adjustment, 2),
            "| Bill:", round(total_bill, 2)
        )
    print("MONTH-TO-MONTH COMPARISON")
    for i in range(1, len(bills)):
        increase = bills[i] - bills[i - 1]
        percent = increase / bills[i - 1] * 100
        print("Month", i + 1, "increase:", round(percent, 2), "%")
        if percent > 30:
            print("Unusual increase detected")
```

8. Test Cases:

Test Case	Input / Condition	Expected Result
1	Energy ₹160; 5% rebate ₹8; fixed ₹50	Bill ₹202.00
2	Energy ₹260; no adjustment; fixed ₹50	Bill ₹310.00
3	Energy ₹900; 5% surcharge ₹45; fixed ₹50	Bill ₹995.00

9. Sample Output Screenshot:

```
Month 1 | Units: 80 | Energy: 160 | Adjustment: -8.0 | Bill: 202.0
Month 2 | Units: 120 | Energy: 260 | Adjustment: 0 | Bill: 310
Month 3 | Units: 190 | Energy: 470 | Adjustment: 0 | Bill: 520
Month 4 | Units: 280 | Energy: 900 | Adjustment: 45.0 | Bill: 995.0
MONTH-TO-MONTH COMPARISON
Month 2 increase: 53.47 %
Unusual increase detected
Month 3 increase: 67.74 %
Unusual increase detected
Month 4 increase: 91.35 %
Unusual increase detected
```

10. Conclusion

The program calculates progressive tariffs, adjustments, and fixed charges, then detects large month-to-month changes using lists, arithmetic, conditions, and loops.